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Data collection is the process of gathering information on the variables of interest on the study. It is done in a planned and systematic manner that that enables research questions to be answers, hypothesis to be tested and to evaluate the results of research findings. Therefore, data collection is a key aspect of the research. Inaccurate data can negatively impact on the analysis and the eventual findings of the research. It becomes important for the researcher to carefully select, design and apply data collection to ensure the quality and integrity of the research.

There are two main types of data collection namely, quantitative and qualitative. The type of data collected from each varies dramatically.

Quantitative data collection methods rely on predetermined sampling and structured data collection instruments. The collected data is easy to summarize, compare and generalize. Qualitative data collection methods play an important role in understanding perceptions and people's *perceived reality*. Qualitative research can often be used prior to a quantitative data collection in helping to generate hypothesis, design the survey and questionnaires.

I. The importance of ensuring integrity in data collection

Accurate data collection is integral to ensuring high quality research. Below is a list of potential dangers if data is not collected properly:

- inability to answer research questions accurately
- inability to repeat and validate the study
- distorted findings resulting in wasted resources
- misleading other researchers to pursue fruitless avenues of investigation
- compromising decisions for industry and society
- causing harm to human participants (think of finance, health and education)
(University of Illinois, 2005)

For IT research the cost of inaccurate research findings can lead to discomfort and harm to areas of society and industry. For example, research into medical diagnoses using image processing or machine learning predictions for investment forecasting needs to be accurate. In addition, the researcher must not falsify data and be guarded against errors. Compromised data collection can lead to dire consequences for the researcher.

2. Quantitative data collection

Quantitative data is usually collected from a large sample. This allows for generalization to a much larger population. Data is numerical (or converted to numbers) which is easily quantifiable. Closed ended questions are often used for this purpose. The *questions* are objective in nature (as opposed to subjective in open-ended questions). Quantitative data can include opinions, perceptions and variables to support or reject a hypothesis, these need to be converted to numerical format for analysis.

Quantitative data can also include images, sound or datasets (e.g. twitter, ITS student records, software product metrics).

Quantitative data collections methods

Quantitative data collections methods are experiments, surveys and interviews.

Quantitative data types

Quantitative data can be descriptive, comparative or relationship. *Descriptive data* allows for a variable of the sample to be quantified. For example:

What is the average time spent in the computer lab per semester for a first year student?

Comparative data allows for a comparison between two or more samples based on one or more variables. For example:

What is the difference in the average time spent in the computer lab per semester between the first year and final year IT student?

Relationship data highlights trends, relationships and associations between two or more variables. For example:

What is the relationship between time spent in the computer laboratory and assessment marks?

Some advantages of quantitative data collection methods

There is less room for researcher subjective judgement and bias because the findings are the result of statistical analysis. Quantitative data is easier to analyse compared to qualitative data. Online survey tools improve the management of collected data. The findings of research from a small sample is generalized to a larger population.

Some disadvantages of qualitative data collection methods

Quantitative data does not cater well for exploratory research. It is designed to answer *what, who and when* type questions and not *how and why* type questions. The sample size has to be large enough to allow for accurate results and generalization to the population. Sometimes finding a large enough sample can present challenges. Confirmation bias is when the researcher deliberately or erroneously leaves out data to favour the hypothesis or research questions.

3. Qualitative data collection

Qualitative data is collected from a smaller group of participants. It is not unusual to have a group of between 10 to 15 participants. Qualitative data is less concerned with numbers and statistics. Instead, it aims to understand behaviours, motivations and feelings of participants. The questions asked are *How* and *Why* which gathers more subjective responses. This is because of the *in-depth* and highly detailed nature of the responses. The analysis of qualitative data is more complex and time consuming.

Qualitative data collections methods

Qualitative data collections methods are interviews, focus groups, observations and expert opinion.

Qualitative data types

Qualitative data exists in the form of *narratives*. These are usually recorded responses that are later analysed. The responses are *rich* and the researcher has to painstakingly analyse for themes and relationships.

Some advantages of qualitative data collection methods

A researcher can pick-up on information through observing facial expressions, body language and tone of voice. These are missed out in an online survey for example. Compared to numerical feedback, the in-depth feedback often provides a better understanding of a participant's rationale for a certain opinion or perspective. There is a smaller pool of participants to manage.

Some disadvantages of qualitative data collection methods

The data analysis for qualitative data is time consuming. The researcher has go through the process of transcribing, sorting, coding and iterative thematic analysis or other techniques. The results are not generalizable due to small sample. Conducting interviews is difficult if the researcher has not spent significant time in developing the necessary skills. Likewise, for interpretation of the data.

4. Survey

A survey is intended to collect data from a large sample of participants. Questionnaires are central to the Survey data collection method. It is a key instrument for quantitative research. Researchers develop, administer and manage surveys and questionnaires using online tools.

Online tools for questionnaires

Questionnaires should be administered electronically. There are rare cases when printed questionnaires may be preferred. Some recent tools to support development of online questionnaires are:

- SoGoSurvey
- QuestionPro
- Surveymonkey
- Google forms
- Survey planet

Formulating questions

The researcher needs to carefully develop the questions for the questionnaire. This usually follows an in-depth literature review to determine the key concepts that the research needs responses to.

For example, should the research aim to determine key practices in a Scrum software project, the researcher would need to develop questions pertaining to knowledge areas (or concepts) in cost estimation, resource allocation, quality, task allocation, requirements gathering, client liaison and tool usage. The responses from participants will provide insights into the practices within each knowledge area.

The **first step** is to determine the type of information the questionnaire intends to gather. Is the questionnaire the best instrument for this purpose? The type of information is derived from the research question.

The type of questions need to be determined **next**. There are generic types listed below:

- Dichotomous: a simple yes/no or agree/disagree type question.
- Open-ended: allows for a response in the participants *own words*. The analysis of responses to this type of question does have its unique set of challenges which the researcher will need to mitigate. This question is useful when trying to determine *how* and *why*.
- Multiple choice: provides the participant with a set of 4 or 5 mutually exclusive responses. This allows for easier analysis of responses. One shortcoming is that the set of responses may not contain the one wanted by the participant.
- Rank-order (ordinal) scale: requires the participant to rank or order responses in a given set. It forces the participant to place importance on some responses over others. All options must be ranked. It does not allow for identifying why or the reasons for this ranking.
- Rating scale: allows for the participant to rate *something* on a given scale with an equal number of positive and negative responses. A popular range is strongly agree, agree, disagree, strongly disagree.

Figures 1 to 3 below provide examples of question types.

Have you ever smoked marijuana?

☐ Yes

☐ No

If yes, about how many times have you smoked marijuana?

☐ Once

☐ 2 to 5 times

☐ 6 to 10 times

☐ 11 to 20 times

☐ more than 20 times

Figure 1: Dichotomous question followed by a logical flow to multiple choice question

What features are important to you when evaluating fitness trackers? Please order the following items from most important to least important.

Measuring Steps	2
Calories Burned	1
Measuring Heartbeat	5
Exercise Tracking	3
Measuring Distance	4

Figure 2: Ranking order type question

* 2. How likely is it that you would recommend this company to a friend or colleague?

NOT AT ALL LIKELY EXTREMELY LIKELY

0	1	2	3	4	5	6	7	8	9	10
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Figure 3: Rating scale type question

The **third** step is to develop the questions. Each question must be clear, concise and direct. There should be no ambiguity or challenges arriving from wording, technical jargon and language. Online tools are used to support the development of questions. Ask one question at a time and carefully consider inclusion of “not applicable” or “not sure” response options. The entire questionnaire should be as short as possible.

Fourth is to pilot test the questionnaire. Before pilot testing, the researcher should approach their supervisor, peer and other experts to evaluate the questionnaire. Following the corrections and adjustment, a small number from the sample can be used for the *pilot testing* of the questionnaire.

5. Focus group

Focus group is usually used as a qualitative data collection technique. There are reports of first use as early as 1940. Traditionally, focus groups were conducted in face-to-face manner however, online tools can be used and they bring some benefits. The focus group brings together 8 to 12 participants (for example, experts in the field) to provide opinion on an artefact (model, system, practice guidelines). The researcher usually acts as the moderator and guides the focus group through the process.

The premise is that more in-depth responses can be collected when a group of participants interact when compared to individual responses. The interaction allows the group to hear the opinions and perspectives of others before passing their own.

The data collected from the focus group is usually qualitative in nature. This is due to the in-depth data provided by a small number of expert participants.

Types of Focus Groups

There are different configurations for a focus group. The following are some popular configurations.

- Two-way focus group - one focus group watches another focus group and discusses the observed interactions and conclusion
- Dual moderator focus group - one moderator ensures the session progresses smoothly, while another ensures that all the topics are covered
- Duelling moderator focus group (fencing-moderator) - two moderators deliberately take opposite sides on the issue under discussion

- Respondent moderator focus group - one and only one of the respondents is asked to act as the moderator temporarily
- Client participant focus groups - one or more client representatives participate in the discussion, either covertly or overtly
- Mini focus groups - groups are composed of four or five members rather than 6 to 12 (Wikipedia, 2021)

Factors for successful focus group

On the surface conducting a focus group might seem simple enough. However, researchers have to ensure certain factors are addressed for an effective focus group. The following factors are reported in Muijeen et al. (2020).

- Good planning and event organization: the participants should be informed early and made well aware of the objectives of the study, allow sufficient time to prepare (researcher and participants), prepare as a moderator, the interaction and questions should be carefully planned as the questions asked *guide* the interaction and the responses.
- Size of the focus groups: a larger group will generate more responses. A smaller group may be used for more complex research artefacts or topics. The researcher needs to decide on the optimum number for generating adequate responses based on the complexity of the session.
- Duration: the duration of the focus group depends on the complexity of the topic or the artefact, the number of questions and the size of the group. However, not more than two hours.
- Environment and location: the venue (room, computer lab) should be comfortable temperature, size and lighting. The seating should be conducive for a focus group usually circular so each participant can see the other.
- Selection of participants: there are two types. *Homogeneous* groups have participants with similar characteristics such as, education and training, work experience and expertise. *Heterogeneous* groups have participants with different characteristics should be used for exploratory purposes. The participants should not be in conflict with each other or with the research.
- Scripts and materials: the interaction should be *scripted* including welcoming statements, introductions to all participants, rules, tasks to be performed during the session.
- Being flexible: the researcher should develop knowledge and skill in open-ended questions, adjusting questions, group conversation and decision theory, pilot study and data collection techniques.
- Interpersonal skills: ability to deal with interruptions, conflict and undesirable behaviour. Personality skills to encourage participation and freedom of speech.
- Non-judgemental and adaptable: the researcher should expect participants to deviate from the topic and the expected responses. These must be taken into account and *collected* as qualitative data. The researcher must not show biasness and attempt to manipulate the responses.

Virtual Focus groups

Virtual focus groups are conducted using an internet based platform as opposed to face-to-face settings. There are two types, synchronous is similar to videoconferencing where participants are free to interact. Asynchronous is similar to a broadcast whereby the researcher addresses the group of participants. Synchronous is the preferred type.

The factors for a successful focus group remain as detailed in the sub-section above. However, there are other considerations for virtual focus groups. Lewis and Muzzy (2020) provide the following guidelines.

- Consider the access to ICT and the reliability of these systems
- The researcher needs to be well versed with the management and setup of these ICTs
- Participants must be allowed time to familiarize themselves with the platform and practice its usage. Video, chat, and virtual hand-raising.
- Consider the use of live camera feeds on the impact of responses and resource usage
- What is the impact of excluding certain experts due to their non-access to ITCs?
- Will the participants have access to a quiet space for the duration of the focus group

Advantages of Focus Groups

Miller (2020) describes the advantages of focus groups.

- A diverse set of responses
- A richer set of responses
- Likeminded group of participants
- Cost effective and able to cross boundaries by using online tools
- Time saving compared to individual interactions
- Allows for the building-on of new concepts that would have not been foreseen by the researcher

Disadvantages of Focus Groups

Miller (2020) describes the disadvantages of focus groups.

- Different level of expertise. Although participants are in the same area or field, their expertise varies from one company to another.
- Individuals may have less speaking time compared to one-on-one interviews or may not be the extrovert
- Strong personalities and outspoken individuals can dominate the group
- The researcher may inadvertently inflict bias upon the group
- Some experts may expect compensation
- Should a participant voluntarily leave the research, the rest of the focus group may be affected marring the expected results

6. Online datasets

Data collection from business and human participants is a long and tedious process. Given the limited time frame and resources in any given research project, the researcher has to explore other sources of reliable data. Online datasets are becoming the norm for access to reliable and large data for IT research. Many IT researchers have abandoned the tedious process of collecting data directly from participants in favour of online datasets.

An online dataset is simply a collection of data available for free or for a small fee. They exist in different formats for example, Excel CSV. Data is collected during previous research of others via online surveys, human interactions or observation or mined from datasources such as websites and tweets. It is vital that the researcher understands for what purpose the dataset was originally created and where the data came from.

Tutorial

1. Would you use quantitative or qualitative data collection for research aimed at determining experiences of first year IT students. Motivate your choice.
2. Would you use quantitative or qualitative data collection for research aimed at determining the best Python developer platforms. Motivate your choice.
3. Differentiate between open-ended and closed-ended questions. Create an open-ended question for question 1 and closed-ended question for question 2.
4. Assume that you are researching the satisfaction of first year IT students. Design three simple Likert scale type question to collect data. Use an online tool.
5. Design a questionnaire with at least one of each type question. The questionnaire should be designed to collect data on the UX of the Google search engine. Use an online tool.
6. List three disadvantages of conducting virtual focus groups.
7. Find a suitable repository of online datasets to conduct research in:
 - a. Machine learning
 - b. Crime prediction

References

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- Miller, B. 2020. 17 Advantages and Disadvantages of a Focus Group. Available: <https://greengarageblog.org/17-advantages-and-disadvantages-of-a-focus-group>. Accessed: 30 January 2021.
- Muijeen, K., Kongvattananon, P. and Somprasert, C. 2020. *The key success factors in focus group discussions with the elderly for novice researchers: a review*. Journal of Health Research, Vol. 34 No. 4, pp. 359-371. <https://doi.org/10.1108/JHR-05-2019-0114>
- Northern University of Illinois. 2005. Responsible conduct in data management. Available: https://ori.hhs.gov/education/products/n_illinois_u/datamanagement/dctopic.html. Accessed: 30 January 2021.

Useful urls

- <https://www.analyticsvidhya.com/blog/2016/11/25-websites-to-find-datasets-for-data-science-projects/>
- <https://www.childtrends.org/publications/conducting-successful-virtual-focus-groups>
- <https://freeonlinesurveys.com/survey-research/qualitative-data-collection>

<https://people.uwec.edu/piercech/researchmethods/data%20collection%20methods/data%20collection%20methods.htm>

<https://programs.online.american.edu/msme/masters-in-measurement-and-evaluation/resources/how-to-conduct-focus-groups-qualitative-research>